# Lecture 6: Generalized Linear Models

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SoE, Westlake University

March 25, 2025

1. Review of Last Week

2. Exponential Families and Generalized Linear Models
   - Motivation
   - Exponential family
   - Application in ML (Generalized Linear Models)

# Reading materials & Reference

**Reading materials:**

- Chapter 3, Stanford CS 229 Lecture Notes, `https://cs229.stanford.edu/notes2022fall/main_notes.pdf`
- Lecture 4, Stanford CS 231n, `http://cs231n.stanford.edu/schedule.html`

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2 Exponential Families and Generalized Linear Models

# Optimal classification for known generating model

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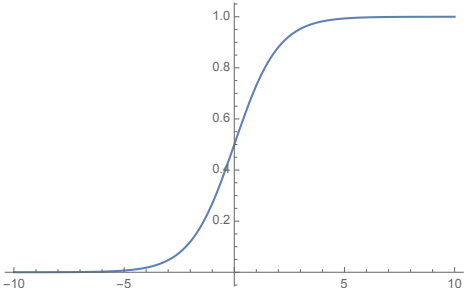
  $\Rightarrow$ Bayes classifier is an unattainable gold standard.
  But we can use the data to learn the distribution (by assuming the data distribution).

# The logistic function

Consider first of all the case of two classes.
The posterior probability for class $\mathcal{C}_1$:

$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \quad (3)$$

$$= \frac{1}{1+\exp(-\eta)} = \sigma(\eta) \quad (4)$$

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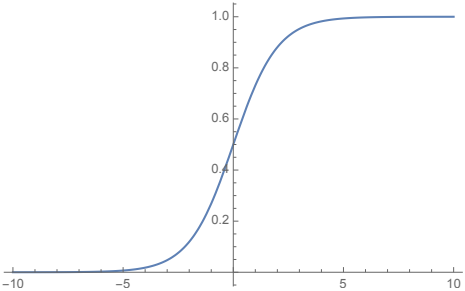
where we have defined

$$\eta = \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \text{ and } \sigma(\eta) := \frac{e^\eta}{1+e^\eta} \tag{5}$$

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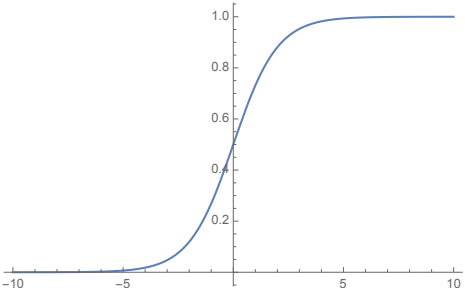


For the case of $K > 2$ classes, we have

$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_j p(\mathbf{x}|\mathcal{C}_j)p(\mathcal{C}_j)} = \frac{\exp(\eta_k)}{\sum_j \exp(\eta_j)} \tag{6}$$

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# Logistic Regression

Given a "new" feature vector $\mathbf{x}$, we predict the (posterior) probability of the two class labels given $\mathbf{x}$ by means of

$$p(1|\mathbf{x}) := \Pr\left[Y = 1|\mathbf{X} = \mathbf{x}\right] = \sigma\left(\mathbf{x}^\top \mathbf{w} + w_0\right) \tag{7}$$

$$p(0|\mathbf{x}) := \Pr\left[Y = 0|\mathbf{X} = \mathbf{x}\right] = 1 - \sigma\left(\mathbf{x}^\top \mathbf{w} + w_0\right) \ , \tag{8}$$

where we predict a real value (a probability) and not a label.

# MLE is a method of estimating the parameters of a statistical model

The MLE finds the parameters $\mathbf{w}^\star$ under which $\{\mathbf{y}, \mathbf{X}\}$ are the most likely:

$$\mathbf{w}^\star = \arg\max_{\mathbf{w}} \left( \mathcal{L}(\mathbf{w}) := \prod_{n=1}^{N} p(\{\mathbf{x}_n, y_n\}|\mathbf{w}) \right) = \arg\min_{\mathbf{w}} \left[ -\log \mathcal{L}(\mathbf{w}) \right] . \tag{9}$$

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The likelihood of the data $\{\mathbf{y}, \mathbf{X}\}$ given the parameter $\mathbf{w}$, i.e., $p(\mathbf{y}, \mathbf{X}|\mathbf{w})$.

$$p(\mathbf{y}, \mathbf{X}|\mathbf{w}) = p(\mathbf{X}|\mathbf{w})p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = p(\mathbf{X})p(\mathbf{y}|\mathbf{X}, \mathbf{w}) , \tag{10}$$

where $\mathbf{X}$ does not depend on $\mathbf{w}$.

# MLE for Logistic Regression

For Logistic Regression, we have:

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \prod_{n=1}^{N} p(y_n|\mathbf{x}_n) \tag{12}$$

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Minimizing $\mathcal{L}(\mathbf{w})$ through the property of stationary points.

$$\nabla\mathcal{L}(\mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{x}_n \left(\sigma(\mathbf{x}_n^\top \mathbf{w}) - y_n\right) = \frac{1}{N}\mathbf{X}^\top \left[\sigma(\mathbf{X}\mathbf{w}) - \mathbf{y}\right] , \tag{13}$$

where $\mathbf{X} \in \mathbb{R}^{N\times d}$.

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where $\mathbf{X} \in \mathbb{R}^{N \times d}$. It has no closed-form solution to $\nabla \mathcal{L}(\mathbf{w}) = 0$.

**Last lecture:**

- Logistic Regression

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**This lecture:**

- Exponential Families
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- **Geometric way:**
  Minimizing the sum of the squares of the residuals:

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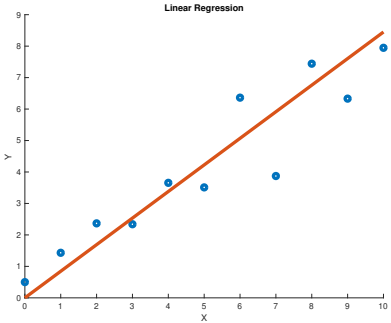

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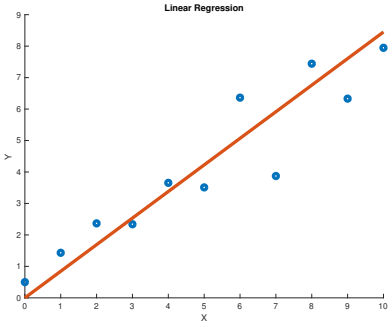

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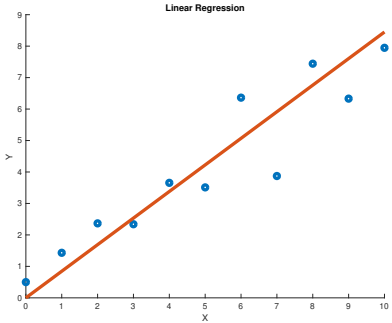
Doing MLE recovers the LS estimator $\hat{\mathbf{w}}$.



# How to get non-linear models?

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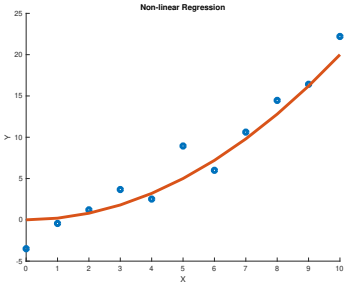
  add non-linear features $(\mathbf{x}, \mathbf{x}^2, \mathbf{x}^3, \ldots)$



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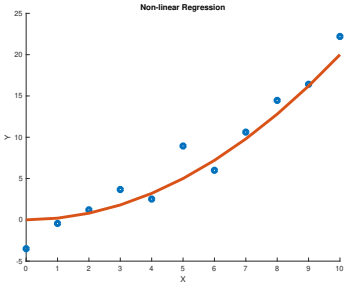

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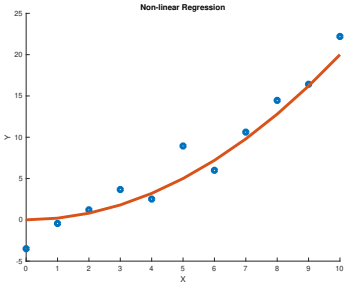

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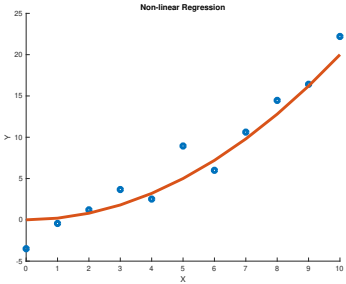

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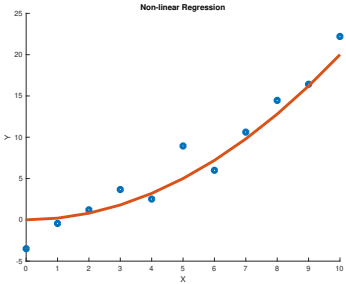

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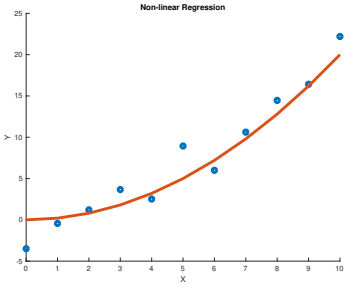

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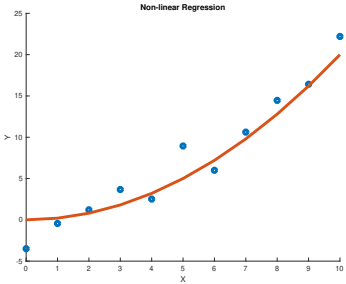


$\Longrightarrow$ Generalized linear model
$\Longrightarrow$ Exponential family

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Logistic Regression models the probability of the two classes $\{0, 1\}$ by

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It is called the **link function**.

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The distribution used in Logistic Regression can be written in a very specific form:

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- The discussion on a class of distributions, known as *exponential families*.
- Many distributions (but not all) fit into this framework and that distributions in this family have many nice properties.

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1 Review of Last Week

2 Exponential Families and Generalized Linear Models
- Motivation
- Exponential family
- Application in ML (Generalized Linear Models)

# Exponential family — definition

A distribution belongs to the exponential family if it can be written in the form

$$p(y|\boldsymbol{\eta}) = \underbrace{h(y)}_{\geq 0} \exp\left[\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) - A(\boldsymbol{\eta})\right] \tag{19}$$

[1] Assume that we are given independent samples from this distribution. We do know $\boldsymbol{\phi}(y)$ and $h(y)$ but not $\boldsymbol{\eta}$. In order to optimally estimate $\boldsymbol{\eta}$ given these samples, all we need is the empirical average of the $\boldsymbol{\phi}(y)$.

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We will exclude such parameters by only looking at the set of parameters

$$\text{Natural parameter space } M := \left\{\boldsymbol{\eta} : \int_y h(y)\exp\left[\boldsymbol{\eta}^\top \boldsymbol{\phi}(y)\right] dy < \infty\right\} \tag{23}$$

# Why?

# Bernoulli distributions belong to the exponential family

**Recall:** A distribution belongs to the exponential family if it can be written in the form

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$\Longrightarrow \eta = g(\mu) = \ln\frac{\mu}{1-\mu} \Leftrightarrow \mu = g^{-1}(\eta) = \frac{e^\eta}{1+e^\eta}$. Link function $g(\mu)$ links the mean of $\phi(y)$ to $\boldsymbol{\eta}$.

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**Link function** ($\eta := g(\mu)$): $\eta_1 = \frac{\mu}{\sigma^2}, \eta_2 = -\frac{1}{2\sigma^2} \iff \mu = -\frac{\eta_1}{2\eta_2}, \sigma^2 = -\frac{1}{2\eta_2}$.

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$$p(y|\boldsymbol{\eta}) = h(y) \exp\left[\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) - A(\boldsymbol{\eta})\right] \tag{33}$$

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# Proof: convexity

- For $\boldsymbol{\eta}_1, \boldsymbol{\eta}_2$ two parameters, we define $\boldsymbol{\eta} = \lambda\boldsymbol{\eta}_1 + (1-\lambda)\boldsymbol{\eta}_2$. We want to show

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- We will continue the proof with the Hoelder's inequality.

# Proof: convexity

- We recall the **Hoelder's inequality:**

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q \tag{44}$$

for $p, q \in [1, +\infty]$ $s.t.$ $\frac{1}{p} + \frac{1}{q} = 1$, and $\|f\|_p = (\int |f(y)|^p \, dy)^{1/p}$.

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- We apply Hoelder's inequality to $f$ and $g$ for $p = 1/\lambda$ and $q = 1/(1-\lambda)$:

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q \quad \text{where} \quad 1/p + 1/q = \lambda + (1-\lambda) = 1 \tag{45}$$

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$$\|g\|_q = \left(\int g(y)^q dy\right)^{1/q} = \left(\int \left[h(y)^{1-\lambda} \exp\left((1-\lambda) \boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right)\right]^{\frac{1}{1-\lambda}} dy\right)^{1-\lambda} = \left(\int h(y) \exp\left(\boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right) dy\right)^{1-\lambda}$$

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- Therefore we have

$$\|f\|_p \, \|g\|_q = \left(\int h(y) \exp\left(\boldsymbol{\eta}_1^\top \boldsymbol{\phi}(y)\right) dy\right)^\lambda \left(\int h(y) \exp\left(\boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right) dy\right)^{1-\lambda} \tag{46}$$

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# Summary of proof: convexity

We have

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$$\leq \left(\int h(y) \exp\left(\boldsymbol{\eta}_1^\top \boldsymbol{\phi}(y)\right) dy\right)^\lambda \left(\int h(y) \exp\left(\boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right) dy\right)^{1-\lambda} \tag{51}$$

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# Derivate of $A(\boldsymbol{\eta})$ and moments: particular cases

- Bernoulli distribution:

$$A'(\boldsymbol{\eta}) = \frac{d}{d\boldsymbol{\eta}} \ln(1 + e^{\boldsymbol{\eta}}) = \frac{e^{\boldsymbol{\eta}}}{1 + e^{\boldsymbol{\eta}}} = \sigma(\boldsymbol{\eta}) = \mu \tag{53}$$

$$A''(\boldsymbol{\eta}) = \frac{d}{\boldsymbol{\eta}} \sigma(\boldsymbol{\eta}) = \sigma(\boldsymbol{\eta}) \left(1 - \sigma(\boldsymbol{\eta})\right) = \mu(1 - \mu) \tag{54}$$

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- Gaussian distribution:

$$\frac{\partial}{\partial \eta_1} A(\boldsymbol{\eta}) = \frac{\partial}{\partial \eta_1} \left( -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2} \ln\left(-\eta_2/\pi\right) \right) = -\frac{\eta_1}{2\eta_2} = \mu \tag{55}$$

$$\frac{\partial}{\partial \eta_2} A(\boldsymbol{\eta}) = \frac{\partial}{\partial \eta_2} \left( -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2} \ln\left(-\eta_2/\pi\right) \right) = \frac{\eta_1^2}{4\eta_2^2} - \frac{1}{2\eta_2} = \mu^2 + \sigma^2 \tag{56}$$

$$\frac{\partial^2}{\partial \eta_1^2} A(\boldsymbol{\eta}) = \frac{\partial}{\partial \eta_1} \left( -\frac{\eta_1}{2\eta_2} \right) = -\frac{1}{2\eta_2} = \sigma^2 \tag{57}$$

# Derivate of $A(\boldsymbol{\eta})$ and moments: general cases

$$\nabla A(\boldsymbol{\eta}) = \nabla \left[ \ln \left( \int h(y) \exp \left( \boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) dy \right) \right] \tag{58}$$

(64)

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# Derivate of $A(\boldsymbol{\eta})$ and moments: general cases

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# Table of Contents

1 Review of Last Week

2 Exponential Families and Generalized Linear Models
- Motivation
- Exponential family
- Application in ML (Generalized Linear Models)

# Maximum Likelihood Estimation (MLE)

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$\Longrightarrow$ The cost function $\mathcal{L}$ is a convex function in $\boldsymbol{\eta}$ since $A(\boldsymbol{\eta})$ is convex.

Given the definition

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Closed-form: assume we have determined the link function $\mathbf{g}(\boldsymbol{\mu}) = \boldsymbol{\eta}$

$$\boldsymbol{\eta} = \mathbf{g}\left(\frac{1}{N}\sum_{n=1}^{N}\boldsymbol{\phi}(y_n)\right) , \tag{70}$$

and justify why we called $\phi(y)$ a sufficient statistics.

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- The GLM framework extends these ideas to the general exponential family.

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We can apply a Generalized Linear Model (GLM)!

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The condition probability is thus modeled as:

$$p(y|\mathbf{x};\mathbf{w}) = h(y_n)\exp\left(\eta\phi(y) - A(\eta)\right) \qquad \text{for} \quad \eta = g \circ f(\mathbf{x}^\top \mathbf{w}) \tag{71}$$

- Two choice points in the specification of a GLM:
  1. The choice of the exponential family distribution
     $\Longrightarrow$ Generally constrained by the nature of the data $Y$
  2. The choice of the response function $f$
     $\Longrightarrow$ Real degree of freedom!
     $\Longrightarrow$ Canonical response function: $f = g^{-1}$, uniquely associated with the given exponential family distribution.

- If we decide to use the canonical response function, the choice of the exponential family density completely determines the GLM:

$$p(y|\mathbf{x};\mathbf{w}) = h(y_n)\exp\left(\eta\phi(y) - A(\eta)\right) \qquad \text{for} \quad \eta = \mathbf{x}^\top\mathbf{w} \tag{72}$$

# Negative log-likelihood estimation

Note that:

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{N}\sum_{n=1}^{N} \ln p(y_n|\mathbf{x}_n^\top \mathbf{w}) = -\frac{1}{N}\sum_{n=1}^{N} (\ln(h(y_n)) + \eta_n \phi(y_n) - A(\eta_n)) \tag{73}$$

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(76)

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$$\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}) = \frac{1}{N}\mathbf{X}^\top \left[\sigma(\mathbf{X}\mathbf{w}) - \mathbf{y}\right] \tag{77}$$

# Some examples

- Gaussian distribution

Least Squares

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- Gaussian distribution — Least Squares
- Bernoulli distribution — Logistic Regression

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- Gaussian distribution — Least Squares
- Bernoulli distribution — Logistic Regression
- Multi-nomial distribution — Softmax Regression

**Last lecture:**

- Logistic Regression

**This lecture:**

- Exponential Families
- Generalized Linear Models